

TAIMOOR UL ISLAM

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Background and Interests

Systems and network architect building the future of broadband and predictable, real-time wireless systems through research, architecture, and real world deployment. With 10+ years of experience and current leadership roles in the NSF and NTIA funded \$19M ARA, \$42M ACCoRD, and \$10M ArMORED projects, I focus on rural broadband, Open RAN, 5G/6G, and scalable network design. I bring strengths in systems and networking architecture, wireless innovation, interoperability, and high impact technical leadership.

Education

Iowa State University

Ph.D. in Computer Engineering (CGPA 3.76/4.00)

May 2026

Ames, Iowa, USA

National University of Sciences and Technology

MS in Electrical Engineering (CGPA 3.75/4.00)

Mar 2020

Islamabad, Pakistan

National University of Sciences and Technology

BE in Electrical Engineering (CGPA 3.32/4.00)

Jun 2014

Islamabad, Pakistan

Selected Publications

- **Taimoor Ul Islam**, Joshua Ofori Boateng, Md Nadim, et al. “Design and Implementation of ARA Wireless Living Lab for Rural Broadband and Applications”, *Computer Networks*, 263 (2025) 111188.
- Guoying Zu, Joshua Ofori Boateng, Varun S. Advani, **Taimoor Ul Islam**, et al. “Real-World Integration and Evaluation of Open-Source 5G Core with Commercial RAN”. In *Proceedings of the 43rd IEEE Military Communications Conference (MILCOM)*, 2025.
- **[Best Paper Award]** Joshua Boateng, Tianyi Zhang, Guoying Zu, **Taimoor Ul Islam**, et al. “AraRACH: Enhancing NextG Random Access Reliability in Programmable Wireless Living Labs”. In *Proceedings of 11th IEEE International Conference on Network Softwarization (IEEE NetSoft 2025)*.
- **[Best Paper Award]** **Taimoor Ul Islam**, Tianyi Zhang, et al. 2023. “AraMIMO: Programmable TVWS mMIMO Living Lab for Rural Wireless”. In *Proceedings of the 17th ACM Workshop on Wireless Network Testbeds, Experimental evaluation & Characterization (WiNTECH)*, 2023.
- **[Best Demo Award]** **Taimoor Ul Islam**, Md, Nadim, Guoying Zu, et al. “ARA PAWR: Enabling Wireless Experiments with Programmable COTS RAN and X-Haul Platforms” presented at *Midscale Experimental Research Infrastructure Forum (MERIF)* 2024.
- Md Nadim, **Taimoor Ul Islam**, Salil Reddy et al. “AraSync: Precision Time Synchronization in Rural Wireless Living Lab”. In *Proceedings of the 30th Annual International Conference on Mobile Computing and Networking (MobiCom 2024)*.
- Tianyi Zhang, Joshua Boateng, **Taimoor Ul Islam**, et al. “ARA-O-RAN: End-to-End Programmable O-RAN Living Lab for Agriculture and Rural Communities”. In *Proceedings of Next-Generation Open and Programmable Radio Access Networks (NG-OPERA)*, *IEEE INFOCOM* 2024.
- Joshua Boateng, Tianyi Zhang, Guoying Zu, **Taimoor Ul Islam**, et al. “AraSDR: End-to-End, Fully-Programmable Living Lab for 5G and Beyond”. In *Proceedings of IEEE International Conference on Communications (ICC)*, 2024.
- **Taimoor Ul Islam**, Joshua Boateng, et al. 2023. “ARA PAWR: Wireless Living Lab for Smart and Connected Rural Communities”. In *Proceedings of the 29th Annual International Conference on Mobile Computing and Networking (ACM MobiCom '23)*. Association for Computing Machinery, New York, NY, USA, Article 98, 1–3. <https://doi.org/10.1145/3570361.3614068>.
- **[Best Paper Honorable Mention Award]** Tianyi Zhang, Guoying Zu, **Taimoor Ul Islam**, et al. “Exploring Wireless Channels in Rural Areas: A Comprehensive Measurement Study” In *Proceedings of IEEE Future Networks World Forum (FNWF)*, 2023.
- **Taimoor Ul Islam**, and F. A. Khan, “Sorted Channel Based User Search for Initial Access in 5G and Beyond Cellular Networks”. In *Proceedings of 2022 IEEE Symposium on Computers and Communications (ISCC)*, Rhodes, Greece, 2022, pp. 1-6, doi: 10.1109/ISCC55528.2022.9912894.
- Kate Keahey, Jason Anderson, Michael Sherman, Cody Hammock, Zhuo Zhen, Jenett Tillotson, Timothy Bargo, Lance Long, **Taimoor Ul Islam**, Sarath Babu, Hongwei Zhang, and François Halbach. 2022. “CHI-in-a-Box: Reducing Operational Costs of Research Testbeds”. In *Practice and Experience in Advanced Research Computing (PEARC '22)*. Association for Computing Machinery, New York, NY, USA, Article 14, 1–8.

Projects

ARA Wireless Living Lab

June 2021 - Present

- Designed, implemented and deployed the \$16M ARA wireless living lab that is currently serving as a testbed for advanced wireless research and also as a living lab to serve rural applications and rural communities.

ACCoRD

Jan 2024 - Present

- Leading the integration and architecture of Open RAN O-CUDU, O-RU and core as part of the \$42M NTIA NOFO-1 funded Acceleration of Compatibility and Commercialization for Open RAN Deployments (ACCoRD) project at the ISU site.

ArMORED

Feb 2025 - Present

- Successfully secured NTIA NOFO-2 funding for \$10M Architecture for mMIMO Open RAN Energy-efficient Devices (ArMORED) project in collaboration with Skylark Wireless and Eridan Inc. Leading the Energy efficiency research study, developing test cases for the new O-RU prototype and deploying and integrating with O-DUCU in field for extensive field measurements.

Work Experience

Iowa State University, WiCI

Jun 2026 – Present

Research Assistant Professor

Ames, Iowa

- Lead Open RAN architecture and integration at the ISU site for the \$42M ACCoRD and \$10M ArMORED programs, owning O-CUDU, O-RU, and core integration along with the ArMORED energy-efficiency research study.
- Direct deployment and field evaluation of Skylark Wireless mMIMO O-RU prototypes (Faros v2 and 32T32R Faros v3), defining test cases and leading multi-cell field measurement campaigns.
- Set the research direction for real-time scheduling and RAN slicing on the ARA living lab, extending it toward precision agriculture connectivity.
- Mentor graduate students and lead the WiCI wireless subgroup across testbed operations and research execution.
- Manage vendor and partner engagements with Skylark Wireless and Eridan on O-RU prototyping, interoperability, and deployment milestones.

Iowa State University, WiCI

Jan 2021 – May 2026

Graduate Research Assistant

Ames, Iowa

- Designed, implemented, and deployed the \$16M ARA wireless living lab, building the end-to-end testbed of routers, switches, servers, radios, antennas, and cabling from planning through operation.
- Led the Skylark Wireless mMIMO deployment on the testbed and used it for dissertation research on URLLC, mMIMO, and predictable rural broadband.
- Created and maintained the public AraMIMO GitLab repository, contributing new PHY and MAC layer APIs to open the mMIMO deployment for community research.
- Deployed an OpenStack-based framework for testbed resource modeling and management, contributing to the project software stack and documentation.
- Coordinated with vendors and contractors to plan and execute multi-site equipment roll-out.

Skylark Wireless, LLC

May 2023 – Aug 2023

Wireless Engineering Intern

Houston, Texas

- Hands-on experience on mMIMO infrastructure of Skylark Wireless and gained knowledge on practical and theoretical aspects of mMIMO wireless communication.
- Worked on writing and refining the modding APIs to expose some of the internal functions for facilitating external collaboration and development of scheduling and grouping algorithms.
- Learned DevOps and CI/CD processes practiced in industry for routine development of software.

National Telecom Corporation

Jul 2015 – Dec 2020

Assistant Divisional Engineer

Islamabad, Pakistan

- Primary role in Planning, Development, Installation, Configuration, Optimization and Troubleshooting of DSLAMs, Switches, Routers, BRAS and other network equipment.
- Ensured seamless connectivity by troubleshooting and resolving IT and network-related issues of customers.
- Planned and forecast internet media of DSLAMs and MSAGs for internet customers based on usage and traffic trends.
- Managed inventory and store for network expansion and faults rectification.
- Supervised customer support staff and ensured 24/7 availability for customer satisfaction.
- Coordinated Wireless broadband (3G/4G) services with third-party operators for subscribers.
- Monitored billing application module for regional customers addressing key issues with application developers and IT department.

Communicators Globe (Pvt) Ltd

Aug 2014 – Jul 2015

Application Support Engineer

Islamabad, Pakistan

- Deployed Cisco Network routers/switches and planned Microwave communication links.
- Trained Customers on Company's products as well as on the technology used.
- Created reports of major milestone activities and documentation of product operation manuals for customers.

Technical Skills

- **Programming Languages:** Python, Matlab, C++
- **Certifications and Training:** CCNA (Route & Switch) 2015, CCNP (Route & Switch) MCSE 2012
- **Networking & Protocols:** TCP/IP, BGP, OSPF, MPLS, VLAN, QoS, SDN, VPN, DHCP, DNS, NAT, IPv4/IPv6
- **Wireless & RAN:** LTE, 5G NR, O-RAN, mMIMO, OpenAirInterface, srsRAN, Skylark Wireless
- **Tools & Platforms:** OpenStack, Docker, Kubernetes, VMware, Ansible, GitLab, CI/CD
- **Hardware & Vendors:** Juniper, Cisco, Huawei, Dell, NI Ettus, Skylark Wireless
- **Monitoring & Security:** PRTG, SolarWinds, Wireshark, SNMP, Firewall configuration, NMS

Recognition and Awards

- ISU Graduate College **Research Excellence Award** 2026.
- ISU Graduate and Professional Student Senate (GPSS) **Research Excellence Award** 2026.
- **Best Paper Award** in proceedings of the 11th IEEE NetSoft 2025.
- **Best Demo Award** at the Midscale Experimental Research Infrastructure Forum (MERIF) 2024.
- **Best Paper Award** in proceedings of the 17th ACM Workshop on Wireless Network Testbeds, Experimental evaluation & Characterization (ACM WiNTECH 2023).
- **Best Paper Honorable Mention Award** in proceedings of the IEEE Future Networks World Forum (IEEE FNWF 2023).
- **L3 Harris Spot Award (2014)** - 2 times for recognition of my hard work and diligence during my work at Communicator's globe (Pvt) Ltd. Islamabad, Pakistan
- **Gold Medal** for best-graduating student in overall academics and extracurricular activities during my Undergraduate studies at NUST, Pakistan.

Professional and Leadership Engagements

- **Student Member Institute of Electrical and Electronics Engineers (IEEE)** (2022 to present) and IEEE NUST student body (2011-2013).
- **Student Member IEEE Communications Society (IEEE ComSoc)** (2022 to present).
- **Student Member Association for Computing Machinery (ACM)** (2021 to present).
- **Joint President** Telecomm Society (2013 - 2014), NUST Pakistan.
- **Prefect Sports and Extra Curricular Activities** (2013 - 2014), NUST Pakistan.
- **Reviewer for Peer Reviewed Journals and Conferences**
 - IEEE Transactions for Wireless Communications.
 - International Journal of Computer and Telecommunications Networking.
 - IEEE Communications Magazine.
 - IEEE International Conference on Computer Communications (INFOCOM).
 - ACM International Symposium on Quality of Service (IWQOS).
 - SN Soft Computing Journal.
 - SN Computer Science Journal.